

Newman-Penrose Formalism Calculations

Stephani et al., Equation (13.55_e-1)

Metric

$$ds^2 = \frac{\cosh(k\tau) e^{\left(2A\tau + \frac{k^2\tau}{2A}\right)}}{k} d\tau \otimes d\tau - \frac{k}{\cosh(k\tau)} dx \otimes dx + \frac{kz}{\cosh(k\tau)} dx \otimes dy + \frac{kz}{\cosh(k\tau)} dy \otimes dx + \left(-\frac{k^2 z^2 - \cosh(k\tau)^2 e^{(2A\tau)}}{k \cosh(k\tau)} \right) dy \otimes dy + \frac{\cosh(k\tau) e^{\left(\frac{k^2\tau}{2A}\right)}}{k} dz \otimes dz$$

Coordinates

$$(x^\mu) = (\tau, x, y, z)$$

Metric Functions

$$X^2 = \frac{\cosh(k\tau)}{k}$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} i \sqrt{2} \sqrt{\frac{\cosh(k\tau)}{k}} e^{\left(A\tau + \frac{k^2\tau}{4A}\right)} \right) d\tau + \left(\frac{1}{2} \sqrt{2} \sqrt{\frac{\cosh(k\tau)}{k}} e^{\left(\frac{k^2\tau}{4A}\right)} \right) dz$$

$$n = \left(\frac{1}{2} i \sqrt{2} \sqrt{\frac{\cosh(k\tau)}{k}} e^{\left(A\tau + \frac{k^2\tau}{4A}\right)} \right) d\tau + \left(-\frac{1}{2} \sqrt{2} \sqrt{\frac{\cosh(k\tau)}{k}} e^{\left(\frac{k^2\tau}{4A}\right)} \right) dz$$

$$m = \left(\frac{i \sqrt{2}}{2 \sqrt{\frac{\cosh(k\tau)}{k}}} \right) dx + \left(\frac{1}{2} i \sqrt{2} \sqrt{\frac{\cosh(k\tau)}{k}} e^{(A\tau)} - \frac{i \sqrt{2} z}{2 \sqrt{\frac{\cosh(k\tau)}{k}}} \right) dy$$

$$\bar{m} = \left(\frac{i\sqrt{2}}{2\sqrt{\frac{\cosh(k\tau)}{k}}} \right) dx + \left(-\frac{1}{2}i\sqrt{2}\sqrt{\frac{\cosh(k\tau)}{k}}e^{(A\tau)} - \frac{i\sqrt{2}z}{2\sqrt{\frac{\cosh(k\tau)}{k}}} \right) dy$$

Contravariant components

$$l = \left(\frac{i\sqrt{2}\sqrt{k}e^{(-A\tau - \frac{k^2\tau}{4A})}}{2\sqrt{\cosh(k\tau)}} \right) d\tau + \left(\frac{\sqrt{2}\sqrt{k}e^{(-\frac{k^2\tau}{4A})}}{2\sqrt{\cosh(k\tau)}} \right) dz$$

$$n = \left(\frac{i\sqrt{2}\sqrt{k}e^{(-A\tau - \frac{k^2\tau}{4A})}}{2\sqrt{\cosh(k\tau)}} \right) d\tau + \left(-\frac{\sqrt{2}\sqrt{k}e^{(-\frac{k^2\tau}{4A})}}{2\sqrt{\cosh(k\tau)}} \right) dz$$

$$m = \left(\frac{i\sqrt{2}\sqrt{k}ze^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}} - \frac{i\sqrt{2}\sqrt{\cosh(k\tau)}}{2\sqrt{k}} \right) dx + \left(\frac{i\sqrt{2}\sqrt{k}e^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}} \right) dy$$

$$\bar{m} = \left(-\frac{i\sqrt{2}\sqrt{k}ze^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}} - \frac{i\sqrt{2}\sqrt{\cosh(k\tau)}}{2\sqrt{k}} \right) dx + \left(-\frac{i\sqrt{2}\sqrt{k}e^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}} \right) dy$$

Directional Derivatives

$$D = \frac{i\sqrt{2}\sqrt{k}e^{(-A\tau - \frac{k^2\tau}{4A})}}{2\sqrt{\cosh(k\tau)}} \partial_\tau + \frac{\sqrt{2}\sqrt{k}e^{(-\frac{k^2\tau}{4A})}}{2\sqrt{\cosh(k\tau)}} \partial_z$$

$$\Delta = \frac{i\sqrt{2}\sqrt{k}e^{(-A\tau - \frac{k^2\tau}{4A})}}{2\sqrt{\cosh(k\tau)}} \partial_\tau - \frac{\sqrt{2}\sqrt{k}e^{(-\frac{k^2\tau}{4A})}}{2\sqrt{\cosh(k\tau)}} \partial_z$$

$$\delta = \frac{i\sqrt{2}\sqrt{k}ze^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}} - \frac{i\sqrt{2}\sqrt{\cosh(k\tau)}}{2\sqrt{k}} \partial_x + \frac{i\sqrt{2}\sqrt{k}e^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}} \partial_y$$

$$\bar{\delta} = -\frac{i\sqrt{2}\sqrt{k}ze^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}} - \frac{i\sqrt{2}\sqrt{\cosh(k\tau)}}{2\sqrt{k}}\partial_x - \frac{i\sqrt{2}\sqrt{k}e^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}}\partial_y$$

Spin Coefficients

$$\rho = -\frac{i\sqrt{2}A\sqrt{k}e^{(-A\tau-\frac{k^2\tau}{4A})}}{4\sqrt{\cosh(k\tau)}}$$

$$\sigma = \frac{i\sqrt{2}A\sqrt{k}e^{(-A\tau-\frac{k^2\tau}{4A})}}{4\sqrt{\cosh(k\tau)}} + \frac{i\sqrt{2}k^{\frac{3}{2}}e^{(-A\tau-\frac{k^2\tau}{4A})}\sinh(k\tau)}{4\cosh(k\tau)^{\frac{3}{2}}} - \frac{\sqrt{2}k^{\frac{3}{2}}e^{(-A\tau-\frac{k^2\tau}{4A})}}{4\cosh(k\tau)^{\frac{3}{2}}}$$

$$\kappa = 0$$

$$\mu = \frac{i\sqrt{2}A\sqrt{k}e^{(-A\tau-\frac{k^2\tau}{4A})}}{4\sqrt{\cosh(k\tau)}}$$

$$\lambda = -\frac{i\sqrt{2}A\sqrt{k}e^{(-A\tau-\frac{k^2\tau}{4A})}}{4\sqrt{\cosh(k\tau)}} - \frac{i\sqrt{2}k^{\frac{3}{2}}e^{(-A\tau-\frac{k^2\tau}{4A})}\sinh(k\tau)}{4\cosh(k\tau)^{\frac{3}{2}}} + \frac{\sqrt{2}k^{\frac{3}{2}}e^{(-A\tau-\frac{k^2\tau}{4A})}}{4\cosh(k\tau)^{\frac{3}{2}}}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{i\sqrt{2}k^{\frac{5}{2}}e^{\left(-A\tau-\frac{k^2\tau}{4A}\right)}}{16A\sqrt{\cosh(k\tau)}} + \frac{i\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau-\frac{k^2\tau}{4A}\right)}\sinh(k\tau)}{8\cosh(k\tau)^{\frac{3}{2}}} - \frac{\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau-\frac{k^2\tau}{4A}\right)}}{8\cosh(k\tau)^{\frac{3}{2}}}$$

$$\gamma = -\frac{i\sqrt{2}k^{\frac{5}{2}}e^{\left(-A\tau-\frac{k^2\tau}{4A}\right)}}{16A\sqrt{\cosh(k\tau)}} - \frac{i\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau-\frac{k^2\tau}{4A}\right)}\sinh(k\tau)}{8\cosh(k\tau)^{\frac{3}{2}}} + \frac{\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau-\frac{k^2\tau}{4A}\right)}}{8\cosh(k\tau)^{\frac{3}{2}}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = \frac{3k^3 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{8 \cosh(k\tau)} + \frac{Ak^2 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)} \sinh(k\tau)}{4 \cosh(k\tau)^2} + \frac{k^4 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)} \sinh(k\tau)}{8A \cosh(k\tau)^2} + \frac{iAk^2 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{4 \cosh(k\tau)^2} + \frac{ik^4 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{8A \cosh(k\tau)^2} + \frac{3ik^3 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)} \sinh(k\tau)}{4 \cosh(k\tau)^3} - \frac{3k^3 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{4 \cosh(k\tau)^3}$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{Ak^2 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)} \sinh(k\tau)}{4 \cosh(k\tau)^2} - \frac{k^3 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)} \sinh(k\tau)^2}{8 \cosh(k\tau)^3} - \frac{iAk^2 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{4 \cosh(k\tau)^2} - \frac{ik^3 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)} \sinh(k\tau)}{4 \cosh(k\tau)^3} + \frac{k^3 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{8 \cosh(k\tau)^3}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{3k^3 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{8 \cosh(k\tau)} + \frac{Ak^2 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)} \sinh(k\tau)}{4 \cosh(k\tau)^2} + \frac{k^4 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)} \sinh(k\tau)}{8A \cosh(k\tau)^2} + \frac{iAk^2 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{4 \cosh(k\tau)^2} + \frac{ik^4 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{8A \cosh(k\tau)^2} + \frac{3ik^3 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)} \sinh(k\tau)}{4 \cosh(k\tau)^3} - \frac{3k^3 e^{\left(-2A\tau - \frac{k^2\tau}{2A}\right)}}{4 \cosh(k\tau)^3}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"